Risk Analyst Report – Anti-Fraud Solution ML

Rudah Lages Nascimento

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1 Data Analysis

1.1 Dataset Overview and Data Integrity

The dataset contains transactional data from a card-not-present (CNP) environment covering a period of two full days. A total of 3,199 transactions were analyzed, with no duplicate `transaction_id` identified, ensuring data consistency and reliability.

The overall fraud (chargeback) rate is **12.22%**, which is significantly high compared to typical benchmarks in the payments industry, indicating a high-risk environment.

Transaction amounts range from **R$1.22 to R$4,097.21**, with an average value of **R$767.81**, suggesting a wide distribution of transaction sizes. Additionally, some `device_id` values are missing, which may impact traceability and should be considered in risk evaluation.

From a behavioral perspective, the maximum number of fraudulent transactions observed was **25 for a single user** and **22 for a single merchant**, indicating the presence of repeated fraudulent activity rather than isolated events.

1.2 Suspicious Behavior Analysis

1.2.1 User Behavior Analysis

The analysis reveals that several users (120) exhibit a **100% fraud rate**, meaning all their transactions resulted in chargebacks. However, an important distinction must be made.

Users with only a single transaction may represent isolated fraud cases or insufficient data. In contrast, users with multiple transactions and a 100% fraud rate are significantly more suspicious, as they indicate consistent fraudulent behavior.

In particular, some users performed multiple transactions (three to four), all resulting in fraud. This pattern strongly suggests intentional fraud activity, potentially involving stolen credentials or synthetic identities. Additionally, users with higher transaction volumes combined with high fraud rates represent the highest-risk profiles.

1.2.2 Device Behavior Analysis

The data shows that each device is associated with a single user, indicating no direct evidence of device sharing. However, this does not imply low fraud risk.

Fraudsters often use techniques such as device spoofing or frequently changing device identifiers to evade detection. This is supported by the presence of users operating across multiple devices.

Several users with **two to three devices exhibit 100% fraud rates**, while one user stands out with **four devices, 31 transactions, and an 80% fraud rate** (Figure 1), indicating sustained and potentially organized fraudulent activity. At the same time, some multi-device users show no fraud, demonstrating that device count alone is not a sufficient indicator and must be evaluated in combination with behavioral patterns.

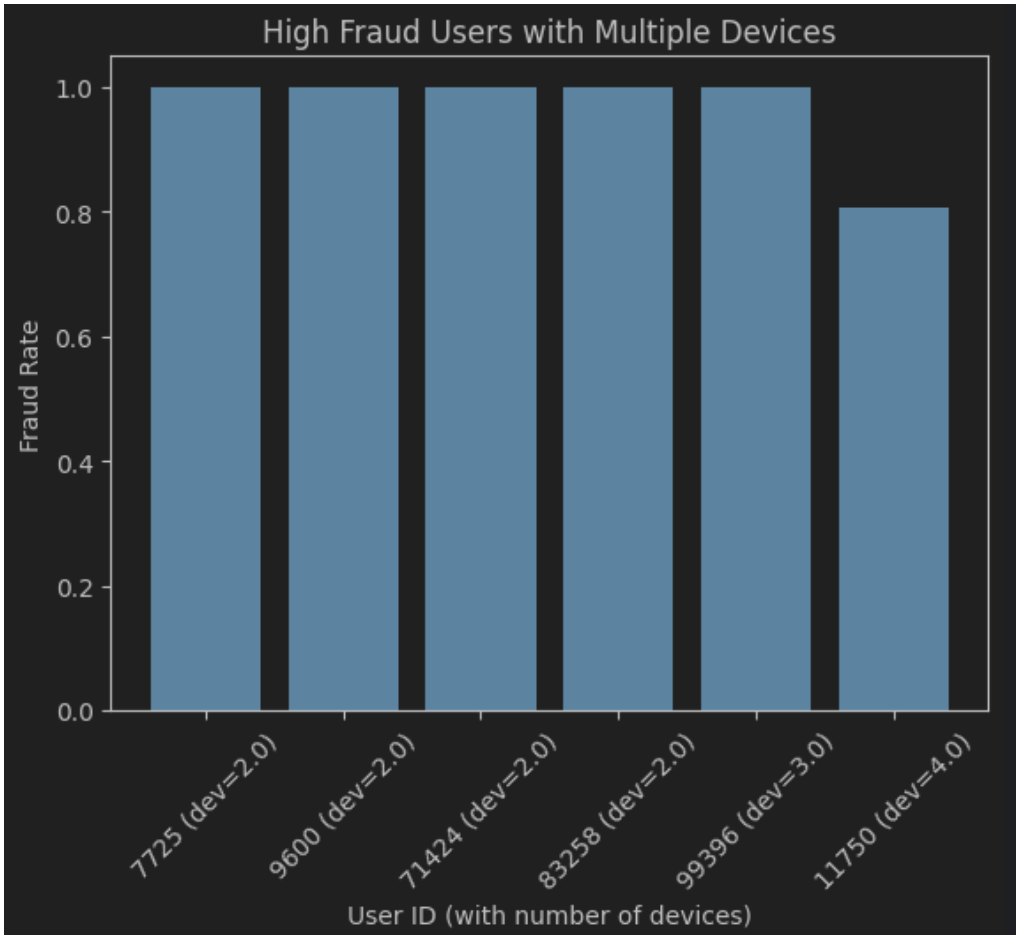

Figure 1 – Users with high fraud rate and multiple devices

Additionally, transactions with available device information show a **higher fraud rate (~13.7%) compared to those without (~8.1%)**, suggesting that device identification alone does not prevent fraud and must be complemented with other signals.

1.2.3 Transaction Amount Behavior Analysis

Fraudulent transactions are strongly associated with higher monetary values. The analysis shows that the average transaction amount for fraud is R$1,453.57, more than twice that of non-fraud (R$672.32), while the median value is R$999.47 compared to R$360.32, nearly three times higher (Figure 2).

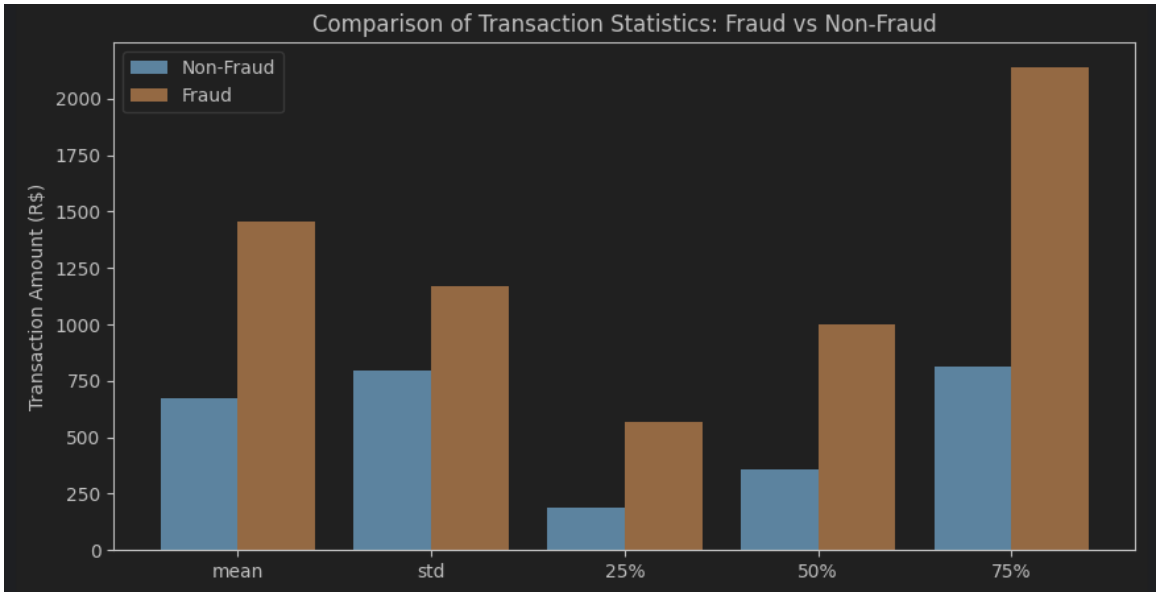

Figure 2 – Comparison of transaction statistics: Fraud x Non-Fraud

This indicates that fraudsters aim to maximize financial gain when executing fraudulent transactions.

1.2.4 Temporal Patterns and Financial Impact

Fraudulent activity exhibits a clear temporal pattern. Higher fraud rates are observed during late-night and early-morning hours, particularly between **19h and 3h**, (with elevated activity also observed at 6 a.m., although it is not statistically significant due to the very low number of transactions, as shown in the subsequent analysis) suggesting activity outside typical user behavior (Figure 3).

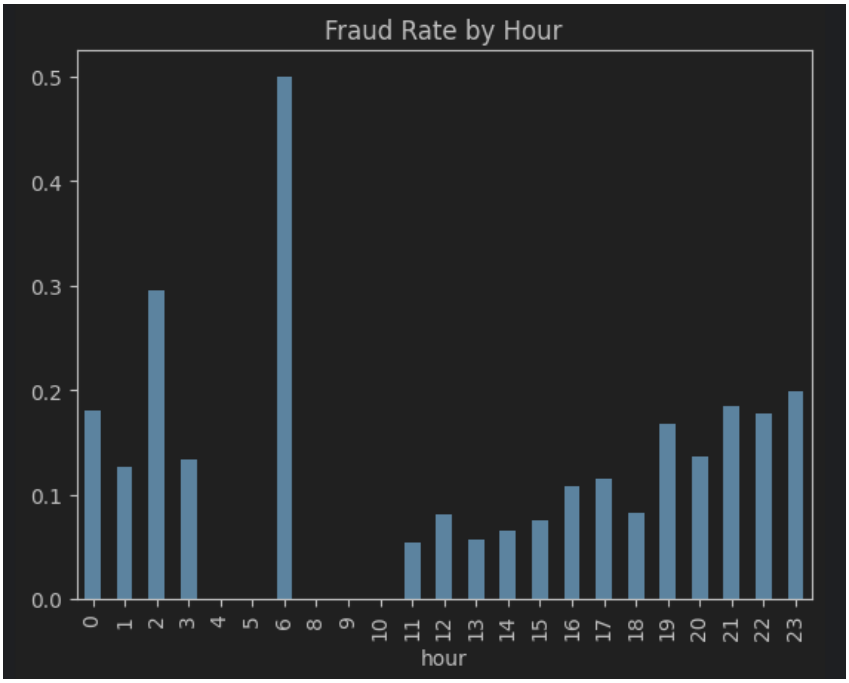

Figure 3 – Fraud Rate by Hour

However, when transaction volume is considered, the most critical risk period shifts to **evening hours (19h–23h)**. During this interval, relatively high fraud rates (approximately 17%–20%) are combined with high transaction volumes, resulting in the largest number of fraudulent transactions (Figure 4).

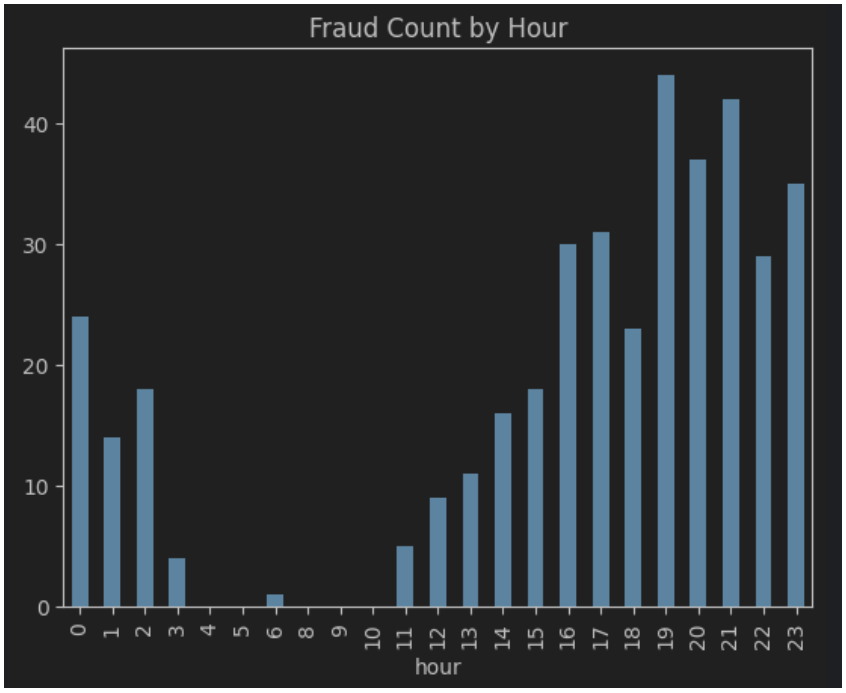

Figure 4 – Fraud Count by Hour

From a financial perspective, fraud impact is highly concentrated between **16h and 23h**, with a peak at **19h**, where fraudulent transactions exceed **R$74,000**. In contrast, early-morning hours, although presenting higher fraud rates, contribute less to total financial loss due to lower transaction volume (Figure 5).

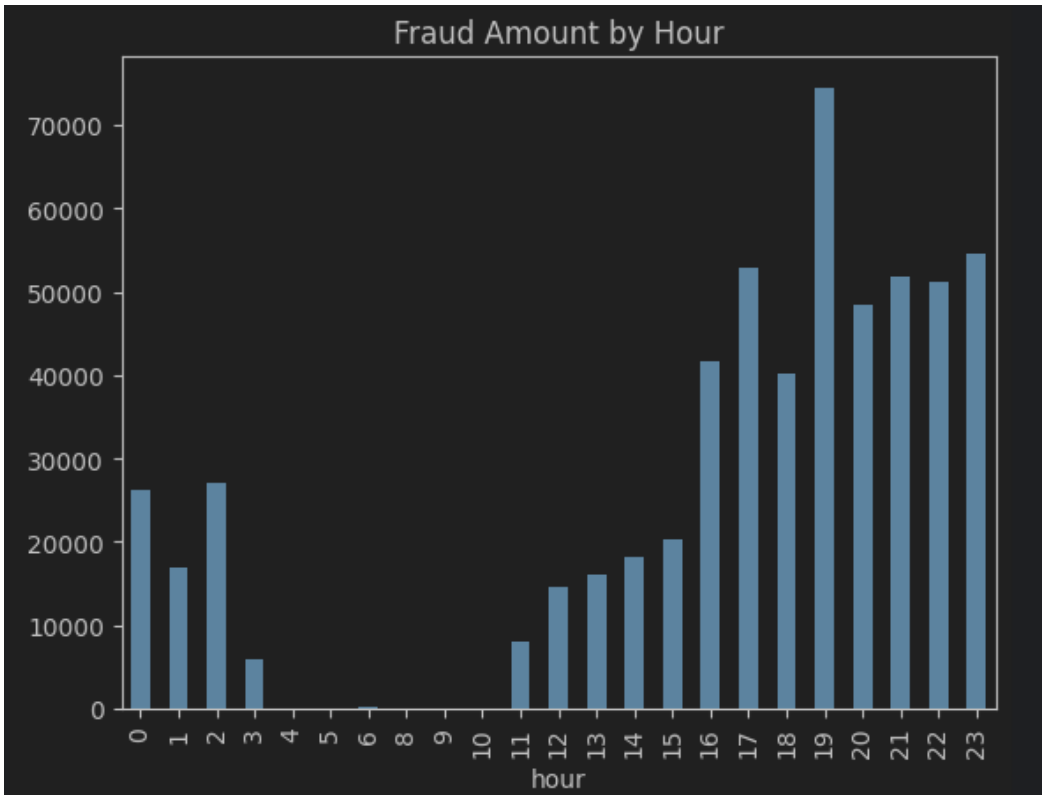

Figure 5 – Fraud Amount by Hour

This demonstrates that fraud risk should be evaluated not only by frequency but also by financial exposure. The most critical periods are those combining high volume, high value, and elevated fraud rates.

1.2.5 Merchant Behavior Analysis

Fraud distribution across merchants is highly uneven and strongly concentrated. A total of **66 merchants exhibit a 100% fraud rate**, meaning all their transactions resulted in chargebacks. Additionally, fraud is heavily concentrated among a small subset of merchants: **only 52 merchants (2.96% of the total) account for 80% of all fraudulent transactions** (Figure 6).

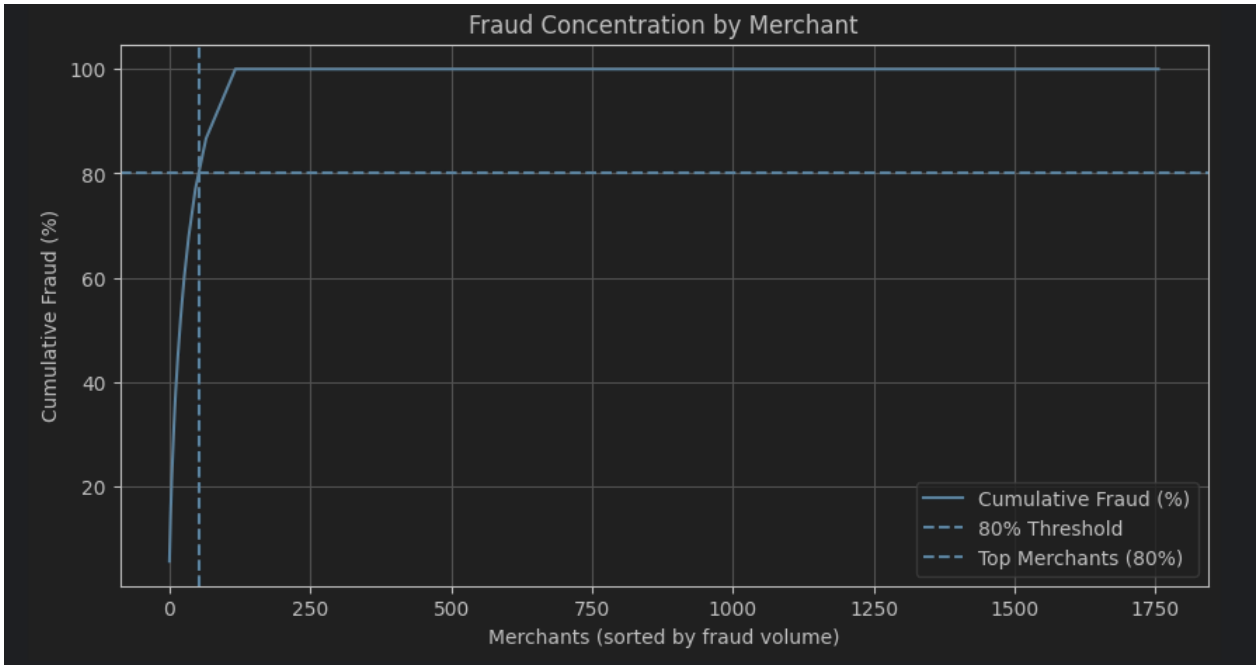

Figure 6 – Fraud concentration by merchant

This indicates that fraud is not randomly distributed across the merchant base, but rather concentrated in specific high-risk entities. While the majority of merchants show little to no fraudulent activity, a small group contributes disproportionately to total fraud and financial losses.

This concentration suggests that targeted monitoring and intervention at the merchant level can significantly reduce overall fraud exposure.

1.2.6 Key Insights

The analysis reveals that fraud in this dataset is not random, but driven by identifiable behavioral and structural patterns.

Key insights include:

- **(i) High Fraud Environment:** The overall fraud rate of 12.22% indicates a significantly elevated risk level compared to typical payment scenarios.

- **(ii) Strong Fraud Concentration:** Fraud is highly concentrated among a small subset of merchants. Only 2.96% of merchants account for 80% of all fraudulent transactions, enabling targeted risk mitigation strategies.

- **(iii) Recurring Fraud Behavior:** Users with multiple transactions and high fraud rates represent the highest-risk profiles, suggesting intentional and repeated fraudulent activity rather than isolated events.

- **(iv) Multi-Device Risk Patterns:** The combination of multiple devices and high fraud rates is a strong indicator of suspicious behavior, although device count alone is not sufficient for detection.

- **(v) High-Value Fraud Strategy:** Fraudulent transactions tend to have significantly higher values, indicating that fraudsters aim to maximize financial gain. Additionally, low-value transactions suggest card testing behavior preceding larger fraud attempts.

- **(vi) Temporal Risk Concentration:** Fraud is more frequent during late-night and evening hours, with the highest operational and financial impact occurring between 19h and 23h.

- **(vii) Financial Impact Concentration:** The largest monetary losses are concentrated in high-volume periods, reinforcing that fraud risk should be evaluated based on financial exposure, not only frequency.

2 Immediate Actions : Rule-Based Risk Control System

Based on the patterns identified in the data analysis, the following actions are recommended to reduce fraud exposure in the short term, following the implementation of a Rule-Based Risk Control System:

2.1 Merchant Risk Control

Rationale: **(ii)**

- Immediately flag and closely monitor the small subset of merchants responsible for the majority of fraudulent transactions.

- Apply stricter approval rules or temporary restrictions for merchants exhibiting very high fraud rates.

- Prioritize manual review or additional verification for transactions associated with these high-risk merchants.

2.2 High-Risk User Monitoring

Rationale: **(iii)**

- Flag users with multiple transactions and high fraud rates, especially those with consistent fraudulent behavior.

- Apply stricter validation or temporarily block users exhibiting repeated fraudulent activity.

- Increase scrutiny on users combining high transaction volume and high fraud rate.

2.3 Multi-Device Behavior Controls

Rationale: **(iv)**

- Increase risk levels for users operating across multiple devices, particularly when combined with suspicious transaction patterns.

- Monitor rapid device changes as a potential signal of device spoofing or fraud attempts.

2.4 Transaction Amount Controls

Rationale: **(v)**

- Apply stricter approval rules or step-up verification for high-value transactions.

- Monitor sequences of low-value transactions followed by high-value transactions as potential card testing behavior.

2.5 Time-Based Risk Adjustments

Rationale: **(vi)**

- Increase risk scoring and monitoring during high-risk periods, particularly between 19h and 23h .

- Apply additional verification steps for transactions occurring during these hours.

2.6 Real-Time Financial Exposure Monitoring

Rationale: **(vii)**

- Prioritize detection and intervention during periods of high transaction volume and high monetary impact.

- Focus on transactions that contribute most to financial losses rather than only high fraud rates.

2.7 Rule-Based Risk Controls System

In the short to medium term, rule-based mechanisms should be implemented to address the most evident risk patterns identified in the analysis.

These include:

- Applying stricter controls for high-risk merchants, especially those responsible for a large share of fraudulent transactions.

- Increasing scrutiny for high-value transactions, particularly above typical transaction ranges.

- Introducing time-based rules, with higher risk levels assigned to transactions occurring during peak fraud hours (19h–23h).

- Flagging suspicious behavioral patterns, such as rapid multi-device usage or repeated failed transactions.

Rule-based systems provide immediate interpretability and are effective for capturing known fraud patterns.

3 Long-Term Recommendations : Anti-Fraud Machine Learning Solution

3.1 Feature Engineering for Fraud Detection

To enable more advanced detection mechanisms, relevant behavioral and transactional features should be engineered based on the insights obtained from the analysis.

Key features include:

- **Transaction-based features:** transaction hour.
- **User-based features:** number of transactions per user, historical fraud rate per user.
- **Device-based features:** number of devices per user.
- **Merchant-based features:** historical fraud rate per Merchant.
- **Behavioral features:** patterns such as sequences of low-value transactions followed by high-value transactions.

These features capture the core fraud signals identified in the dataset and serve as the foundation for predictive modeling.

3.2 Machine Learning-Based Risk Scoring

A supervised machine learning model can be implemented to estimate the probability of a transaction being fraudulent.

A baseline model such as **Logistic Regression** or **Random Forest** is sufficient to capture key patterns and provide interpretable results.

The model would:

- Use the engineered features to predict fraud probability.
- Output a **risk score** for each transaction.
- Continuously improve as new labeled data becomes available.

This approach allows the system to detect complex interactions between variables that are not easily captured by static rules.

3.3 Decision Engine

The output of the model should be integrated into a decision framework that classifies transactions into actionable categories:

- **Approve:** low-risk transactions.
- **Review:** medium-risk transactions requiring manual verification.
- **Reject:** high-risk transactions with strong fraud indicators.

This layered decision process ensures a balance between fraud prevention and customer experience.

4 Additional Data for Fraud Detection

While the dataset provides valuable insights into transactional behavior, additional data sources would significantly enhance fraud detection capabilities by adding contextual and behavioral depth.

In particular, incorporating information such as account age would help distinguish between established and newly created users, which are often associated with higher fraud risk. Network-level data, including VPN or proxy detection, would improve the identification of attempts to mask user location. Payment-related attributes, such as card type and the relationship between the issuing country and the transaction location, could reveal inconsistencies commonly linked to fraudulent activity.

Additionally, behavioral signals such as transaction velocity—multiple transactions within short time intervals—and patterns of repeated or failed attempts would provide strong indicators of suspicious activity. These types of data allow for the detection of abnormal sequences of behavior, which are often more indicative of fraud than isolated transactions.

Overall, the integration of these additional data sources would enable a more comprehensive and context-aware fraud detection framework.

5 Machine Learning and Decision Engine Implementation

The full implementation of the analysis and machine learning model is available in the following repository:

GitHub Repository:

https://github.com/RudahNascimento/fraud-detection-analysis.git

The repository includes the complete code, data processing steps, and model development pipeline.

5.1 Model Performance

The model outputs probability scores rather than binary predictions, enabling flexible risk-based decision making.

The implementation demonstrated solid performance in distinguishing fraudulent from non-fraudulent transactions (Figures 7 and 8).

```
              precision    recall  f1-score   support

       False       0.95      0.98      0.97       843
        True       0.86      0.66      0.74       117

    accuracy                           0.94       960
   macro avg       0.90      0.82      0.86       960
weighted avg       0.94      0.94      0.94       960

Approve    870
Reject      68
Review      22
```

Figure 7 – Model Classification Report and Decision Engine Count

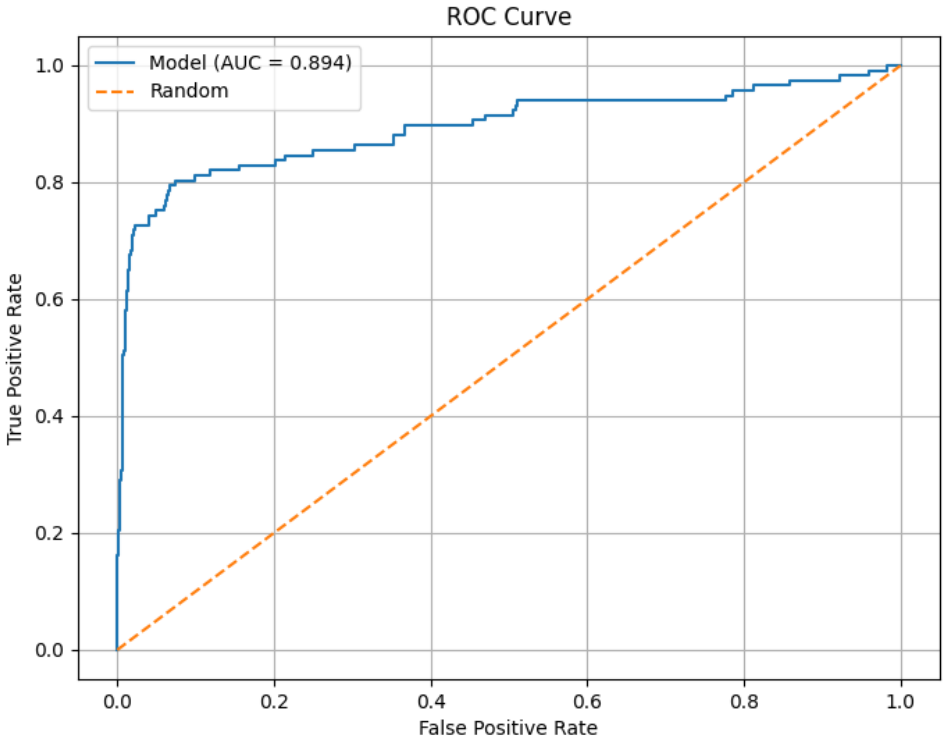

Figure 8 – ROC Curve Plot

These results indicate that the model has good discriminatory power, although there is a clear trade-off between precision and recall.

This trade-off reflects a fundamental balance in fraud detection systems: increasing recall allows the model to capture more fraudulent transactions, reducing financial losses, while higher precision helps minimize false positives, preserving customer experience. In other words, improving fraud detection often comes at the cost of potentially impacting legitimate users.

To address this, the decision threshold was adjusted to prioritize recall, ensuring that a larger portion of fraudulent transactions is detected, while maintaining acceptable precision levels.

5.2 Decision Engine Output

The model's probability output was integrated into a decision engine to classify transactions into actionable categories.

This distribution shown in Figure 8 indicates that most transactions are classified as low-risk, while a smaller portion is flagged for rejection or manual review.

The relatively small review segment reflects the model's strong confidence in its predictions, as most transactions fall into clearly defined risk categories.

5.3 Key Insights from the Model

The machine learning model reinforces several findings from the exploratory analysis:

- Fraud is strongly associated with behavioral and transactional patterns.
- High-value transactions and specific time windows increase fraud risk.
- Merchant and user historical behavior are highly predictive features.

These results validate the importance of combining data analysis with predictive modeling in fraud detection systems.

5.4 Opportunities for Improvement

Although the proposed model provides a solid baseline for fraud detection, several enhancements could be implemented to further improve performance and robustness.

Different algorithms, such as Random Forest or other ensemble methods, could be explored and compared to identify models with better predictive performance and the ability to capture non-linear relationships in the data. Additionally, applying cross-validation techniques such as K-Fold would provide a more reliable estimate of model performance, reducing the risk of overfitting and improving generalization.

Hyperparameter tuning, using approaches such as Grid Search or Random Search, could also be employed to optimize model parameters and extract better performance from the selected algorithm.

Finally, techniques to address class imbalance, such as smoothing or resampling methods, could be applied to improve the model's ability to detect fraudulent transactions, particularly by increasing recall without excessively compromising precision.

6 Understand the Industry Questions

6.1 Question 01

In a typical payment flow, when a transaction is initiated, payment data is sent from the merchant to the processor and then to the acquirer, which routes the request through the card network to the issuing bank. The issuer evaluates the transaction and returns an approval or rejection. While the information flow happens instantly during authorization, the money flow occurs later during settlement, when funds move from the issuing bank to the acquirer and ultimately to the merchant. Each player has a specific role: the merchant initiates the transaction, the acquirer processes it, the network routes it, and the issuer authorizes it.

6.2 Question 02

The acquirer is the entity directly connected to card networks and responsible for processing and settling transactions for merchants. A sub-acquirer acts as an intermediary, managing multiple merchants under its structure while relying on a primary acquirer for processing. A payment gateway is a technological interface that captures and securely transmits payment data from the merchant to the acquirer or processor. The flow remains conceptually the same, but the presence of a gateway or sub-acquirer introduces additional layers between the merchant and the acquirer, often adding services such as risk management, onboarding, and transaction orchestration.

6.3 Question 03

A chargeback occurs when a cardholder disputes a transaction through the issuing bank, resulting in a forced reversal of funds after settlement. This differs from a cancellation, which is initiated by the merchant before settlement and typically avoids penalties. In the acquiring ecosystem, chargebacks are closely linked to fraud, especially in card-not-present environments, as they often represent unauthorized transactions or customer disputes related to fraudulent activity.

6.4 Question 04

An anti-fraud system is a mechanism that evaluates the risk of a transaction before authorization, typically using a combination of rules, behavioral analysis, and machine learning to generate a risk score. Acquirers use anti-fraud systems to reduce fraud-related losses, control chargeback rates, and protect merchants, integrating these solutions into the transaction flow to enable real-time approval, rejection, or manual review decisions.